

Clue Chain Hunt — Technical Report

Inter IIT Bootcamp Phase 2 · Hardware PS · IIT Guwahati

1. System overview

Node / file	Package	Role
hunt_node	clue_hunt_solver	leader autonomy: perception, clue logic, Nav2 client
follower_node	clue_hunt_solver	follower autonomy (camera + wheel odometry only)
hunt.launch.py	clue_hunt_solver	single command that starts both nodes (use_sim_time:=True)
sim.launch.py	clue_hunt_gazebo	Gazebo Fortress, both robots, ros_gz_bridge
navigation.launch.py	clue_hunt_navigation	map_server + AMCL + Nav2 + RViz2 on the saved map

Data flow. Leader: /camera/image_raw, /scan, TF → hunt_node → Nav2 navigate_to_pose → /cmd_vel. Follower: /follower/camera/image_raw, follower/odom TF → follower_node → /follower/cmd_vel. Published: /hunt/clues, /hunt/boards, /hunt/treasure (+ /leader/status). The leader sets its own AMCL initial pose at (0,0) (it spawns at the origin).

2. Clue detection, pose estimation, frames

- ArUco:** DICT_4X4_50, side 0.24 m; wrapper supports both the old and new OpenCV ArUco API.
- solvePnP:** solvePnPGeneric with IPPE_SQUARE and ITERATIVE; the candidate with the lowest reprojection error and positive depth is kept. Poses behind the camera, farther than 5.5 m, or facing away (flipped PnP solution) are discarded.
- Frames:** marker → cam_optical_link → (TF2, at the image timestamp) → map. The board frame (+X out of the face, +Y reader's right, +Z up) is related to the OpenCV marker frame by a fixed rotation R_MB. Each board keeps all observations; its position and normal are the **median of the 10 closest views** (close views are the most accurate). Board position is published on /hunt/boards as id x y.
- QR decoding:** the region 0.30 m right of the marker is projected into the image with the estimated pose, cropped, upscaled (1×/2×/3×) and decoded with cv2.QRCodeDetector, with and without CLAHE.
- Validation:** a clue is HUNT:<n>:<token>:<command>; it is valid only if n equals the expected index and token equals SHA-1(previous clue text)[:4]. The first token is derived from START.

3. Clue-solving and search strategy

- Wait for Nav2, the map and the camera; publish the initial pose.
- For board n: if a candidate with id n is already seen, approach it; otherwise go to the hinted area, **spin 360°**, then visit a **ring of 6 waypoints** (r = 1.3 m) around the hint; if still not found, run a **grid exploration** over free map cells (greedy nearest-neighbour tour).
- Approach:** drive to a stand-off (1.2 m, then 0.9 m and 1.6 m as retries) along the board normal, face the board, re-estimate from close range, decode the QR.
- Reject:** decoy boards are ignored by id; look-alike boards (right id, wrong token) are rejected and the next nearest candidate is tried.
- Commands:** GOTO x y → search around the point; REL a b → point relative to the board frame; PILLAR <colour> → pillar located at run time; BETWEEN A B f → point at fraction f between two pillars; TREASURE prefix → drive onto the point, publish /hunt/treasure, log DONE.
- Pillar finder:** HSV colour blob in the camera gives the bearing; the LiDAR range at that bearing (+0.2 m radius) gives the centre; observations are clustered with a median and outlier rejection. Nothing is hard-coded.
- Goals are snapped to the nearest free cell of the inflated map (unknown cells count as blocked).

4. Follower design

- Detects **tag 49** (0.12 m) in the follower camera; leader centre = tag position 0.21 m ahead along the tag normal; transformed to follower/odom (wheel odometry only).
- Breadcrumbs:** leader positions are stored every 0.12 m; the follower runs **pure pursuit on the trail** (lookahead 0.7 m), so it follows the leader's path rather than cutting corners into walls.
- Speed:** leader speed estimate (low-passed feed-forward) + P-term on the gap (kp_gap 1.5) around d_des = 1.2 m; stops at d_stop = 0.9 m while still facing the leader; speed is scaled by cos(heading error) on turns.
- Tag lost:** keeps following crumbs for 3 s, then spins to re-acquire. It uses no data from the leader (no LiDAR, map or leader topics except the status string /leader/status).